

8)Write a Java program to implement a simple producer-consumer problem using wait() and notify().

PROGRAM:

```
class SharedData {  
  
    int value;  
  
    boolean available = false;  
  
    synchronized void produce(int value) {  
        while (available) {  
            try {  
                wait();  
            } catch (InterruptedException e) {  
                System.out.println(e);  
            }  
        }  
  
        this.value = value;  
        available = true;  
        System.out.println("Produced: " + value);  
  
        notify();  
    }  
  
    synchronized void consume() {  
        while (!available) {  
            try {  
                wait();  
            }  
        }  
    }  
}
```

```
    } catch (InterruptedException e) {  
        System.out.println(e);  
    }  
}
```

```
System.out.println("Consumed: " + value);  
available = false;
```

```
    notify();  
}  
}
```

```
class Producer extends Thread {
```

```
    SharedData data;
```

```
    Producer(SharedData data) {
```

```
        this.data = data;
```

```
    }
```

```
    public void run() {
```

```
        for (int i = 1; i <= 5; i++) {
```

```
            data.produce(i);
```

```
        }
```

```
    }
```

```
}
```

```
class Consumer extends Thread {
```

```
    SharedData data;
```

```

Consumer(SharedData data) {
    this.data = data;
}

public void run() {
    for (int i = 1; i <= 5; i++) {
        data.consume();
    }
}
}

public class Main {
    public static void main(String[] args) {

        SharedData data = new SharedData();

        Producer p = new Producer(data);
        Consumer c = new Consumer(data);

        p.start();
        c.start();
    }
}

```

OUTPUT:

Produced: 1

Consumed: 1

Produced: 2

Consumed: 2

Produced: 3

Consumed: 3

Produced: 4

Consumed: 4

Produced: 5

Consumed: 5
